

PE-EC701B — Satellite Communication

Semester VII • 3L:0T:0P • 3 Credits

Module 5 Exam Notes: Satellite Link Budget

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Power Flux Density and Received Signal Power

The link budget calculations track the signal power level from the ground transmitter, through the orbital space path, to the ground receiver.

1.1 Power Flux Density (F)

An isotropic radiator distributes its power uniformly in all directions over a sphere. At a distance d from the transmitter, the power flux density (F) is:

$$F = \frac{P_t}{4\pi d^2} \quad [\text{W/m}^2] \quad (1)$$

If the transmitter uses a directional antenna with gain G_t , the power is concentrated, yielding:

$$F = \frac{P_t G_t}{4\pi d^2} = \frac{EIRP}{4\pi d^2} \quad [\text{W/m}^2] \quad (2)$$

Where $EIRP = P_t G_t$ is the **Equivalent Isotropically Radiated Power**.

1.2 Received Signal Power (P_r)

The receiving antenna intercepts a portion of this flux. The captured power is proportional to the antenna's **effective aperture area** (A_e):

$$P_r = F \cdot A_e \quad [\text{W}] \quad (3)$$

Using the fundamental antenna relation $A_e = \frac{G_r \lambda^2}{4\pi}$:

$$P_r = \left(\frac{P_t G_t}{4\pi d^2} \right) \left(\frac{G_r \lambda^2}{4\pi} \right) = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad [\text{W}] \quad (4)$$

This is the **Friis Transmission Equation**. It can be written as:

$$P_r = \frac{EIRP \cdot G_r}{L_{FSL}} \quad (5)$$

Where L_{FSL} is the **Free Space Path Loss (FSL)**:

$$L_{FSL} = \left(\frac{4\pi d}{\lambda} \right)^2 = \left(\frac{4\pi f d}{c} \right)^2 \quad (6)$$

In decibels:

$$L_{FSL_dB} = 32.44 + 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}}) \quad (7)$$

2. System Noise Temperature and Noise Power

The performance of a receiving system is limited by thermal noise, which is modeled using an equivalent noise temperature.

2.1 Noise Power Calculation

The thermal noise power (N) generated in a receiver of bandwidth B is given by:

$$N = k \cdot T_s \cdot B \quad [\text{W}] \quad (8)$$

Where:

- $k = 1.38 \times 10^{-23}$ J/K (Boltzmann's constant)
- T_s = System Noise Temperature (Kelvin, K)
- B = Noise Bandwidth (Hz)

In dBW:

$$N_{\text{dBW}} = -228.6 + 10 \log_{10}(T_s) + 10 \log_{10}(B) \quad (9)$$

2.2 Calculation of System Noise Temperature (T_s)

If an antenna with noise temperature T_{ant} is connected to a receiver via a lossy waveguide line with loss factor L ($L \geq 1$, where $L = 10^{\text{Loss}_{\text{dB}}/10}$) at physical temperature $T_0 \approx 290$ K, the system noise temperature (T_s) referred to the receiver input is:

$$T_s = \frac{T_{ant}}{L} + T_0 \left(1 - \frac{1}{L}\right) + T_{rx} \quad (10)$$

Where T_{rx} is the effective noise temperature of the receiver itself, related to its noise figure F_{rx} by:

$$T_{rx} = T_0(F_{rx} - 1) \quad (11)$$

For a cascaded chain of amplifiers (e.g., LNA followed by mixer and down-converter), the overall receiver noise temperature is given by **Friis Formula**:

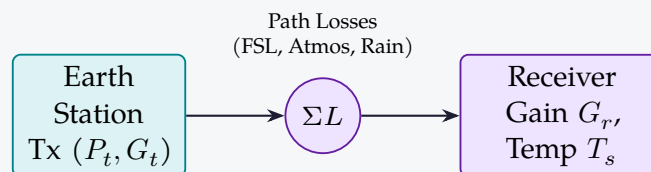
$$T_{rx} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \dots \quad (12)$$

Where T_n and G_n are the equivalent noise temperature and linear gain of the n -th stage. This formula proves why a high-gain, low-noise LNA must be the first stage.

3. Drafting the Satellite Link Budget

The link budget is a tabular summation of all gains and losses along the communication path to calculate the final **Carrier-to-Noise Ratio** (C/N) at the receiver demodulator.

Link Budget System Model



3.1 Carrier-to-Noise Ratio (C/N) Expression

The received carrier power is $C = P_r$. The carrier-to-noise ratio is:

$$\frac{C}{N} = \frac{P_r}{kT_s B} = \frac{EIRP \cdot G_r}{L_{\text{losses}} \cdot kT_s B} = \left(\frac{EIRP}{L_{\text{losses}} \cdot k \cdot B} \right) \left(\frac{G_r}{T_s} \right) \quad (13)$$

Where $G/T = G_r/T_s$ is the receiving system's **Figure of Merit** (expressed in dB/K).

In logarithmic decibel form:

$$\left[\frac{C}{N} \right]_{\text{dB}} = EIRP_{\text{dBW}} - L_{\text{losses_dB}} + \left[\frac{G}{T} \right]_{\text{dB/K}} - k_{\text{dBW}} - B_{\text{dB-Hz}} \quad (14)$$

Where:

- $k_{\text{dBW}} = -228.6 \text{ dBW}/(\text{K} \cdot \text{Hz})$
- $B_{\text{dB-Hz}} = 10 \log_{10}(B_{\text{Hz}})$
- $L_{\text{losses_dB}} = L_{\text{FSL}} + L_{\text{atmospheric}} + L_{\text{polarization}} + L_{\text{antenna-pointing}}$

3.2 Carrier-to-Noise Density Ratio (C/N₀)

To evaluate system performance independent of receiver bandwidth, we use C/N_0 :

$$\frac{C}{N_0} = \frac{C}{N} \cdot B \quad [\text{Hz}] \quad (15)$$

$$\left[\frac{C}{N_0} \right]_{\text{dB-Hz}} = EIRP_{\text{dBW}} - L_{\text{losses_dB}} + \left[\frac{G}{T} \right]_{\text{dB/K}} + 228.6 \quad (16)$$

4. Link Performance: Clear Air vs. Rainy Conditions

Atmospheric phenomena significantly degrade link performance, especially at frequencies above 10 GHz (Ku and Ka bands).

4.1 Clear Air Conditions

Under clear skies, the link is subject only to **gaseous absorption** from water vapor and oxygen molecules in the atmosphere. The attenuation is low (typically < 0.5 dB for C-band).

4.2 Rainy Conditions

Rain introduces a dual penalty on the link:

1. **Signal Attenuation (A_{rain}):** Rain droplets scatter and absorb the microwave energy, reducing the received carrier power (C) by A_{rain} dB.
2. **Increase in Noise Temperature (ΔT_{ant}):** The absorbing rain medium acts as a thermal radiator, increasing the antenna noise temperature:

$$T_{\text{ant_rain}} = T_{\text{ant_clear}} \cdot 10^{-A_{\text{rain}}/10} + T_m (1 - 10^{-A_{\text{rain}}/10}) \quad (17)$$

Where T_m is the physical temperature of the rain medium (typically assumed to be 275 K or 290 K).

This increase in T_{ant} directly raises the system noise temperature T_s , thereby raising the noise floor (N). This combined effect severely degrades the overall received $[C/N]$ ratio.

Quick Revision — Module V

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Key Formulas at a Glance

Equivalent Radiated Power	$EIRP = P_t G_t \Rightarrow EIRP_{\text{dBW}} = P_{t\text{dBW}} + G_{t\text{dBi}}$
Free Space Path Loss	$L_{FSL} = \left(\frac{4\pi d}{\lambda}\right)^2 \Rightarrow L_{FSL\text{dB}} \approx 32.44 + 20 \log(d_{\text{km}}) + 20 \log(f_{\text{MHz}})$
Noise Power	$N = kT_s B \Rightarrow N_{\text{dBW}} = -228.6 + 10 \log(T_s) + 10 \log(B)$
Friis Cascade Temperature	$T_{rx} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \dots$
Carrier-to-Noise Ratio	$\left[\frac{C}{N}\right]_{\text{dB}} = EIRP_{\text{dBW}} - L_{\text{losses_dB}} + \left[\frac{G}{T}\right]_{\text{dB/K}} - 10 \log(B) + 228.6$

One-Line Definitions

- **EIRP:** Equivalent power that would need to be radiated by an isotropic antenna to achieve the same directional signal strength.
- **Free Space Loss:** Geometrical spreading loss of electromagnetic energy as it radiates outward through space.
- **G/T Ratio:** Figure of merit of a receiving system, representing the ratio of antenna gain to system noise temperature.
- **Friis Formula:** Expression calculating the total noise temperature of cascaded receiver stages.
- **Boltzmann's Constant (k):** Proportionality constant relating kinetic energy to temperature (1.38×10^{-23} J/K).
- **Rain Fade Penalty:** Combined loss of carrier power and rise in receiver noise floor caused by rain.

Frequently Asked / Viva Questions

Q1. Define EIRP and write its equation in logarithmic form.

Equivalent Isotropically Radiated Power. It measures the total directional power radiated. $EIRP_{\text{dBW}} = P_{t\text{dBW}} + G_{t\text{dBi}}$.

Q2. What is Free Space Path Loss (FSL) and why does it occur?

It is the attenuation of a wave due to the spherical spreading of its energy over distance. $L_{FSL} = (4\pi d/\lambda)^2$. It does not represent energy absorption by matter.

Frequently Asked / Viva Questions (contd.)

Q3. Explain the physical meaning of the G/T ratio.

It is the receiver's Figure of Merit. A higher G/T indicates a highly sensitive receiving station (high gain G or low system noise temperature T).

Q4. How does the first stage of a receiver cascade affect its noise performance?

According to Friis Formula ($T_{rx} = T_1 + T_2/G_1 + \dots$), if the first stage has high gain (G_1), it divides and minimizes the noise contributions of all subsequent stages (T_2, T_3). Thus, the first stage dictates the overall noise figure.

Q5. Calculate the noise power (in dBW) for a 36 MHz transponder bandwidth at a system temperature of 400 K.

$$N = -228.6 + 10 \log(400) + 10 \log(36 \times 10^6) = -228.6 + 26.02 + 75.56 = -127.02 \text{ dBW.}$$

Q6. What is the difference between C/N and C/N_0 ?

C/N is the carrier-to-noise ratio within a specific bandwidth B . C/N_0 is the carrier-to-noise density ratio, normalized to a 1 Hz bandwidth ($C/N_0 = (C/N) \cdot B$).

Q7. Explain the dual penalty rain inflicts on satellite links.

Rain absorbs and scatters the carrier signal, reducing C . Simultaneously, the absorbing rain emits thermal noise, increasing the antenna temperature and raising the noise floor N .

Q8. An Earth station has a dish antenna with 40 dBi gain and an LNA with 100 K noise temperature. The antenna noise temperature is 50 K under clear skies. Calculate the clear-sky G/T .

$$T_s = T_{ant} + T_{rx} = 50 + 100 = 150 \text{ K. } G/T = 40 - 10 \log_{10}(150) = 40 - 21.76 = 18.24 \text{ dB/K.}$$

Q9. What is the physical temperature T_0 used in noise figure definitions and what is its value?

It is the standard reference temperature for noise measurements, defined as 290 K (17°C).

Q10. Why are link margins drafted into satellite designs?

A link margin (typically 3 to 6 dB) is added to ensure the system remains operational during atmospheric degradation, rain fade, or antenna pointing errors.

Q11. Write the relation between Noise Figure (F) and equivalent noise temperature (T_e).

$$T_e = T_0(F - 1) \text{ where } T_0 = 290 \text{ K.}$$

Q12. How does the slant path impact rain attenuation in satellite links?

A lower elevation angle increases the slant path length through the rain cell, which increases the total rain attenuation compared to a vertical path.

Drafting a Sample Link Budget Table

Link Parameter	Value (Clear Air)	Value (Rainy - 3 dB fade)
Transmitter Power (P_t)	10 dBW	10 dBW
Transmit Antenna Gain (G_t)	45 dBi	45 dBi
EIRP	55 dBW	55 dBW
Free Space Loss (L_{FSL})	−200 dB	−200 dB
Atmospheric/Rain Loss (L_a)	−0.5 dB	−3.5 dB (includes rain)
Receiver Antenna Gain (G_r)	35 dBi	35 dBi
System Noise Temp (T_s)	150 K (21.8 dB-K)	250 K (24.0 dB-K) (increases due to rain)
Boltzmann Constant (k)	−228.6 dBW/Hz-K	−228.6 dBW/Hz-K
C/N₀ Ratio	96.3 dB-Hz	91.1 dB-Hz